



PROJECT SKYNET INITIATIVE

Sunnyvale Headquarters, CA

TECHNICAL ARCHITECTURE BRIEFING

NETWORK PROJECT PROTOTYPE #1

Comprehensive architecture blueprint for autonomous network backbones, encrypted mesh relays, and zero-trust security layers.

PREPARED BY

Network Engineering Division

TARGET FACILITY

Strategic Operations Command

TEAM MEMBERS:

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Julian Low





PHYSICAL INFRASTRUCTURE

HARDWARE SPECIFICATIONS (REQ. 1)

COMPUTE PROCESSING

DC/NPS/LibreNMS Server:
1x Lenovo P350 Workstation

CBR and SYD Workstations:
2x Lenovo P350 Workstation

MULTI-LAYER SWITCHING

CBR-SW1:
1x Cisco Catalyst 9800L 24 4x1G

SYD-SW1 & 2:
2x Cisco Catalyst 9800L 24 4x1G

ROUTING

CBR-GW:
Cisco Catalyst C8200-1N-4T

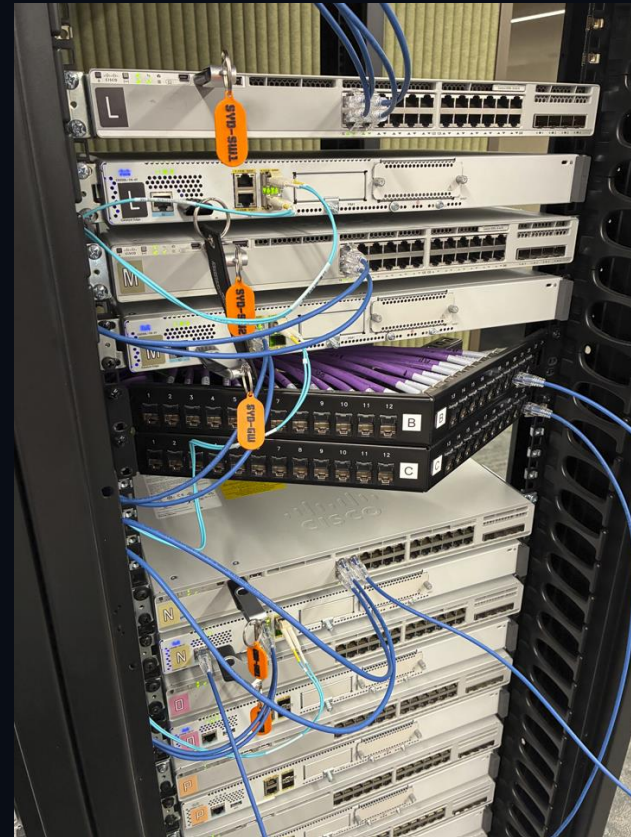
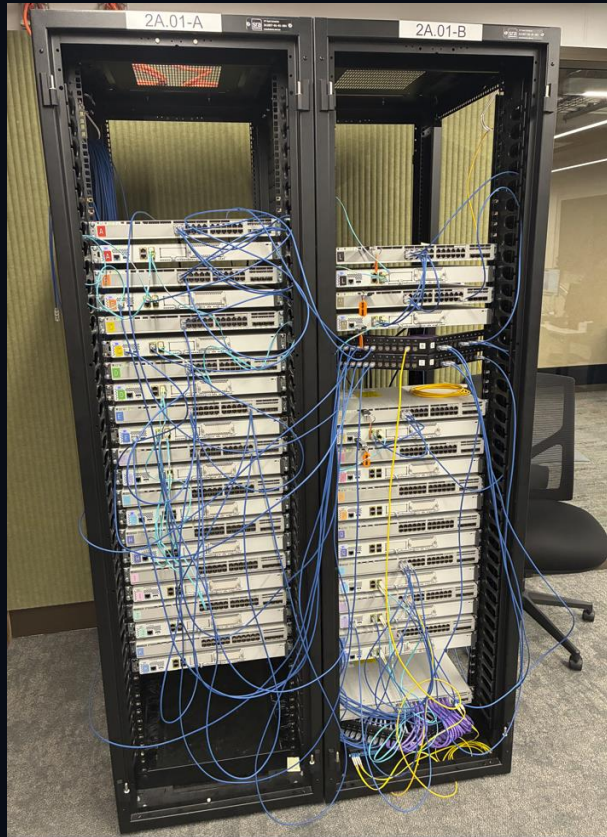
ISP:
Cisco Catalyst C8200-1N-4T

SYD-GW:
Cisco Catalyst C8200-1N-4T



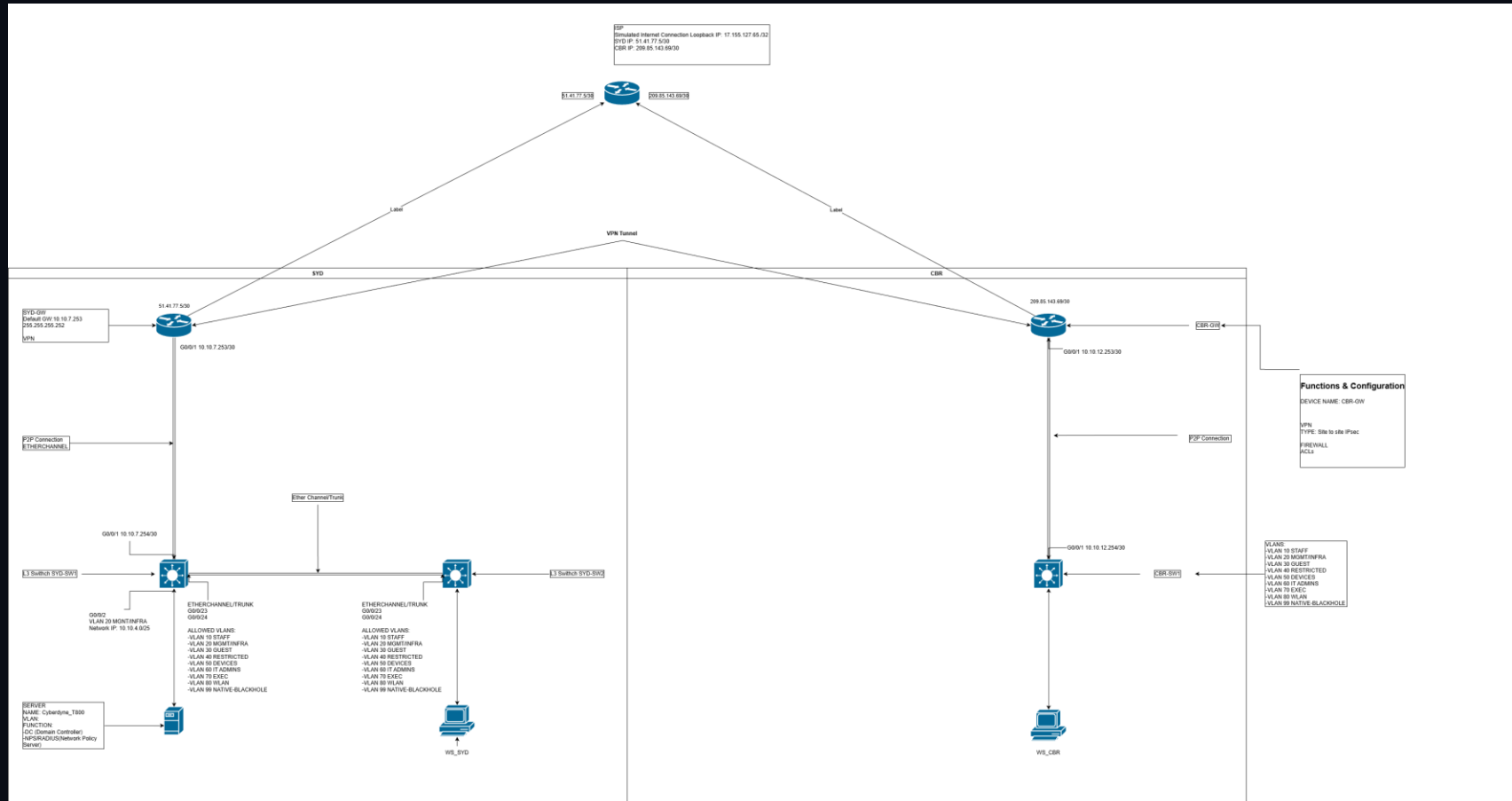
PHYSICAL INFRASTRUCTURE

HARDWARE SPECIFICATIONS (REQ. 1)



TACTICAL BACKBONE

NETWORK DESIGN AND ADDRESSING (REQ. 2)





ENCRYPTED TRUNKING

VPN TUNNEL ARCHITECTURE (REQ. 3)

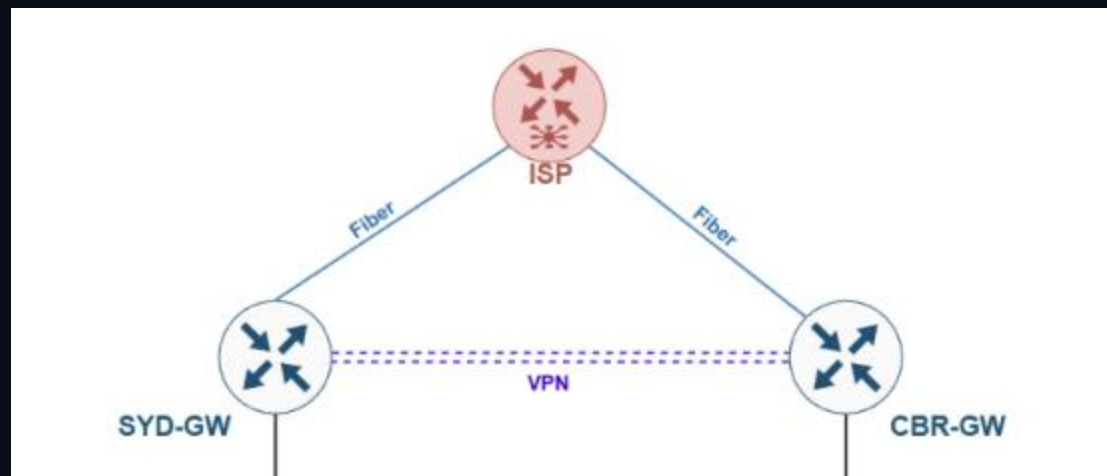
SITE-TO-SITE IPSEC TUNNEL

Logical GRE tunnel protected by IPsec between SYD-GW and CBR-GW

Encapsulating Security Payload (ESP) encryption protocol set to Advanced Encryption Standard (AES) with a strong 256-bit key length.

Hash-based Message Authentication Code (HMAC) with **SHA-256** for data integrity and authentication.

Transport mode is the best-practice design choice when overlaying IPsec on top of GRE tunnels or VTI (Virtual Tunnel Interfaces)



NETWORK PROTOCOLS

IP ADDRESSING PLAN (REQ. 4)

COOPER'S MAGICAL IP ADDRESSING MACHINE

VLAN ID	VLAN / Network	Purpose	Allocate Subnet?	DHCP?	Current Hosts	25% Growth	DHCP Reserved Addresses	Required Hosts	CIDR	Usable Hosts	Spare Hosts	Subnet Mask	Block Size	VLSM Order	Network Address	Broadcast Address	First Usable Host	Last Usable Host	Default Gateway	Network/CIDR	DHCP Start	DHCP End	DHCP Pool Size
SYDNEY (SYD) – VLAN & VLSM ADDRESSING																							
10	STAFF	Staff Endpoints	Yes	Yes	650	162.5	0	813	/22	1022	208	255.255.252.0	1024	1	10.10.0.0	10.10.3.255	10.10.0.1	10.10.3.254	10.10.0.1	10.10.0.0/22	10.10.0.2	10.10.3.254	1021
20	SERVER-INFRA	Servers & Infrastructure	Yes	No	50	12.5	0	63	/25	126	62	255.255.255.128	128	2	10.10.4.0	10.10.4.127	10.10.4.1	10.10.4.126	10.10.4.1	10.10.4.0/25			
30	GUEST	Guest Devices	Yes	Yes	80	20	0	100	/25	126	25	255.255.255.128	128	3	10.10.4.128	10.10.4.255	10.10.4.129	10.10.4.254	10.10.4.129	10.10.4.128/25	10.10.4.130	10.10.4.254	125
40	RESTRICTED	Restricted Endpoints	Yes	Yes	80	20	0	100	/25	126	25	255.255.255.128	128	4	10.10.5.0	10.10.5.127	10.10.5.1	10.10.5.126	10.10.5.1	10.10.5.0/25	10.10.5.2	10.10.5.126	125
50	DEVICES	Printers / IoT / devices	Yes	Yes	45	11.25	0	57	/26	62	4	255.255.255.192	64	5	10.10.5.128	10.10.5.191	10.10.5.129	10.10.5.190	10.10.5.129	10.10.5.128/26	10.10.5.130	10.10.5.190	61
60	ITADMIN	IT Admin Endpoints	Yes	Yes	35	8.75	0	44	/26	62	17	255.255.255.192	64	6	10.10.5.192	10.10.5.255	10.10.5.193	10.10.5.254	10.10.5.193	10.10.5.192/26	10.10.5.194	10.10.5.254	61
70	EXEC	Executive Endpoints	Yes	Yes	24	6	0	30	/26	62	31	255.255.255.192	64	7	10.10.6.0	10.10.6.63	10.10.6.1	10.10.6.62	10.10.6.1	10.10.6.0/26	10.10.6.2	10.10.6.62	61
99	MANAGEMENT	Network Infrastruture	Yes	No	3	0.75	0	4	/29	6	1	255.255.255.248	8	8	10.10.6.64	10.10.6.71	10.10.6.65	10.10.6.70	10.10.6.65	10.10.6.64/29			
998	NATIVE	Unused Native VLAN	No	No		0	0	0															
999	BLACKHOLE	Unused ports VLAN	No	No		0	0	0															
CANBERRA (CBR) – VLAN & VLSM ADDRESSING																							
10	STAFF	Staff Endpoints	Yes	Yes	42	10.5	0	53	/26	62	8	255.255.255.192	64	1	10.20.0.0	10.20.0.63	10.20.0.1	10.20.0.62	10.20.0.1	10.20.0.0/26	10.20.0.2	10.20.0.62	61
20	SERVER-INFRA	Servers & Infrastructure	Yes	No	10	2.5	0	13	/28	14	0	255.255.255.240	16	5	10.20.0.224	10.20.0.239	10.20.0.225	10.20.0.238	10.20.0.225	10.20.0.224/28			
30	GUEST	Guest Devices	Yes	Yes	24	6	0	30	/26	62	31	255.255.255.192	64	2	10.20.0.64	10.20.0.127	10.20.0.65	10.20.0.126	10.20.0.65	10.20.0.64/26	10.20.0.66	10.20.0.126	61
40	RESTRICTED	Restricted Endpoints	Yes	Yes	24	6	0	30	/26	62	31	255.255.255.192	64	3	10.20.0.128	10.20.0.191	10.20.0.129	10.20.0.190	10.20.0.129	10.20.0.128/26	10.20.0.130	10.20.0.190	61
50	DEVICES	Printers / IoT / devices	Yes	Yes	12	3	0	15	/27	30	14	255.255.255.224	32	4	10.20.0.192	10.20.0.223	10.20.0.193	10.20.0.222	10.20.0.193	10.20.0.192/27	10.20.0.194	10.20.0.222	29
60	IT-ADMIN	IT Admin Endpoints	Yes	Yes	3	0.75	0	4	/29	6	1	255.255.255.248	8	7	10.20.1.0	10.20.1.7	10.20.1.1	10.20.1.6	10.20.1.1	10.20.1.0/29	10.20.1.2	10.20.1.6	5
70	EXEC	Executive Endpoints	Yes	Yes	8	2	0	10	/28	14	3	255.255.255.240	16	6	10.20.0.240	10.20.0.255	10.20.0.241	10.20.0.254	10.20.0.241	10.20.0.240/28	10.20.0.242	10.20.0.254	13
99	MANAGEMENT	Network Infrastruture	Yes	No	3	0.75	0	4	/29	6	1	255.255.255.248	8	8	10.20.1.8	10.20.1.15	10.20.1.9	10.20.1.14	10.20.1.9	10.20.1.8/29			
998	NATIVE	Unused Native VLAN	No	No		0	0	0															
999	BLACKHOLE	Unused ports VLAN	No	No		0	0	0															



LOGICAL ISOLATION

VLAN DESIGN & SEGMENTATION (REQ. 5)

VLAN 10

STAFF

Regular users

VLAN 20

INFRASTRUCTURE

Servers and Infrastructure

VLAN 30

GUESTS

Guest Network

VLAN 40

RESTRICTED

For failed logins

VLAN 50

DEVICES

Printers and IoT Devices

VLAN 60

IT ADMIN

System and Infrastructure Admins

VLAN 70

EXECUTIVE

Executive Management

VLAN 99

MANAGEMENT

For Remote Device Management

VLAN 998

NATIVE

Unused Native VLAN

VLAN 999

BLACKHOLE

For unused ports

INTER-VLAN ROUTING

Configured via SVIs on Multi-layer Switches





DEFENSE IN DEPTH

SECURITY ARCHITECTURE DESIGN (REQ. 6)

100%

Zero-Trust Enforcement

DEVICE HARDENING

All intermediary devices have privileged password configured with username/password authentication with all plaintext passwords encrypted.

SECURE REMOTE MANAGEMENT

Via SSHv2, only authorised hosts (IT-Admins) allowed.

SNMP SERVER

LibreNMS Server tracks Health, Performance and Traffic of Routers, Switches and Other Devices across Network



SWITCHING PROTECTION

SECURITY - LAYER 2 CONTROLS (REQ. 7)

Etherchannel

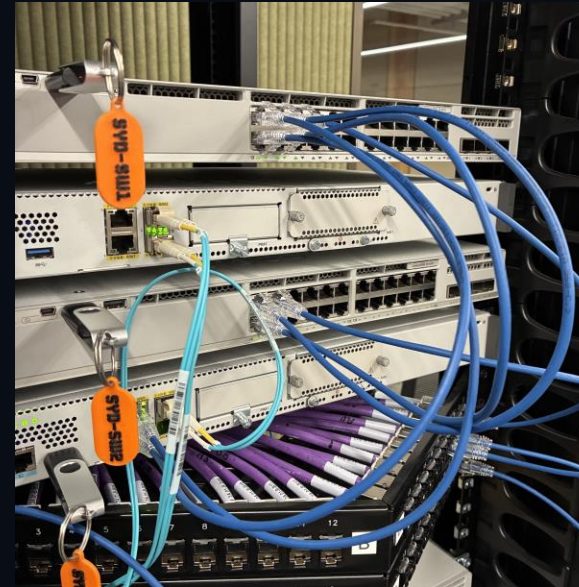
Port-channel links between all Gateway Routers and Switches

802.1x Authentication (LIVE DEMO)

Computers placed in a restricted VLAN until the computer and user is authenticated, at which time they are moved to the Staff VLAN

Port-Security

Violation level: RESTRICT



Etherchannel links



ROUTING PROTECTION

SECURITY - LAYER 3 FILTERING (REQ. 8)

ACCESS CONTROL LISTS

To be demonstrated in the 'Testing' section

Hosts on the Restricted VLAN have no access outside that VLAN

Hosts on the Guest VLAN have no access to internal resources but should be able to access

Hosts on the Staff VLAN have access to internal server resources (at both sites) and Web, Email and FTP servers on the internet. They have no access to the Restricted, Guest or Administration networks

Hosts on the IT Staff and Infrastructure VLANs should have complete access to the rest of the network. They also have access to Web, Email, FTP and SSH servers on the internet.

Implicit deny in ACLs blocks unauthorized internet access

Network Project 1

ACL Documentation

Group 3

Network Project 1

Access Control List Design and Justification

Working reference prepared after physical ACL implementation and testing

1. Design Overview

The Layer 3 ACL design uses inbound filtering on source VLAN SVIs. This means traffic is evaluated as it leaves its originating VLAN, before it is routed toward another VLAN, the remote site, or the Internet.

- Extended ACLs are applied inbound to the source VLAN SVI.
- Rules are ordered from specific exceptions and required services to **broader denies**.
- An explicit final deny is used where a VLAN is intentionally restricted.
- DHCP is permitted before restrictive **rules** so clients can still obtain centrally issued leases from CBR-GW.
- DNS and NTP are explicitly permitted where the policy **requires** access to Cyberdyne_T800 at 10.10.4.10.
- Management access is handled separately using a standard ACL intended for VTY access-class filtering, not as an inbound ACL on VLAN99.
- VLAN20 SERVER-INFRA and VLAN50 DEVICES are intentionally left without inbound SVI ACLs at this stage. Access to them is controlled by source VLAN policies.

Cisco IOS ACLs are stateless. A restrictive inbound ACL on a server or device VLAN can unintentionally block legitimate reply traffic. For this reason, the project primarily controls access from the initiating/source VLAN.

2. VLAN and Subnet Reference

VLAN	Sydney	Canberra	Role
10 STAFF	10.10.0.0/22	10.20.0.0/26	User network
20 SERVER-INFRA	10.10.4.0/25	10.20.0.224/28	Servers and infrastructure
30 GUEST	10.10.4.128/25	10.20.0.64/26	Guest access
40 RESTRICTED	10.10.5.0/25	10.20.0.128/26	Restricted / unauthenticated
50 DEVICES	10.10.5.128/26	10.20.0.192/27	Shared devices / printers / IoT
60 ITADMIN	10.10.5.192/26	10.20.1.0/29	IT administration
70 EXEC	10.10.6.0/26	10.20.0.240/28	Executive users
99 MANAGEMENT	10.10.6.64/29	10.20.1.8/29	Network management

3. High-Level Policy Matrix

Policy	Allowed	Denied	Reason
RESTRICTED	DHCP only for routed traffic	All routed traffic outside its VLAN	Assessment requirement: isolated network
GUEST	DHCP, DNS/NTP to server, web and email Internet services	All other internal resources and unapproved traffic	Internet access without internal network access
STAFF	Permitted internal resources, ICMP, web, email, FTP	Guest, Restricted, ITADMIN and Management at both sites	Assessment-defined staff access policy
ITADMIN	Full internal access, ICMP, web, email, FTP, SSH	Anything not explicitly allowed	Administrative role requires broad network access

A. Cooper

M. Cook

J. Low



INFRASTRUCTURE SERVICES

NETWORK SERVICES & INTEGRATION (REQ. 9)



NTP configured on AD/NPS

Atomic clock synchronization for all subnets.



DNS Server on AD/NPS

Encrypted DNS-over-HTTPS with localized memory cache.



DHCP Server configured on CBR-GW

High-availability IP address allocation mesh.



IIS Web Server on AD/NPS

Host and manage websites, web applications, and APIs.



LibreNMS Server

Centrally collect, monitor, and manage data from networked devices like routers, switches, and servers.

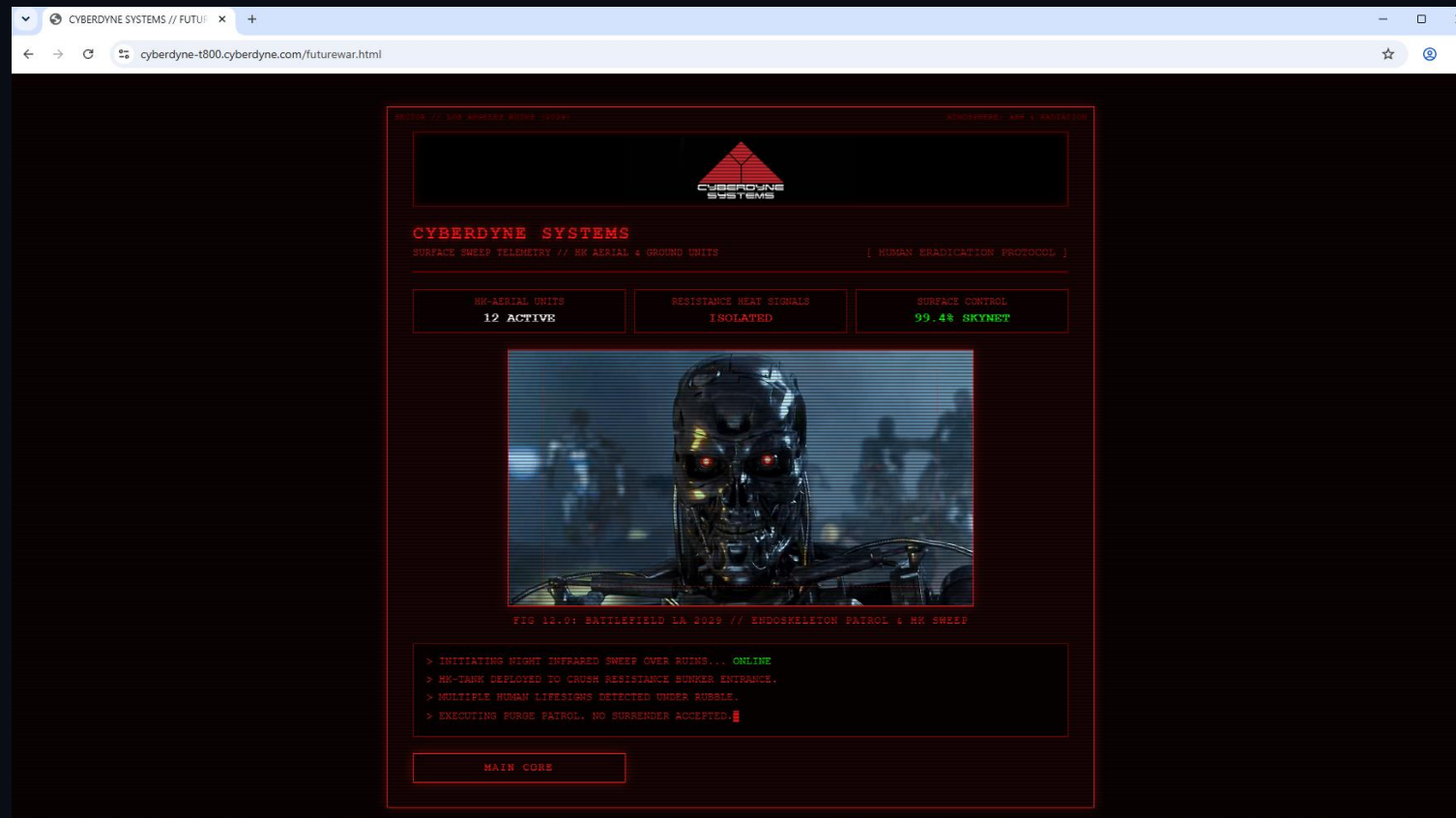




PERIMETER ARCHITECTURE

INTERNET & INTRANET SIMULATION (REQ. 10)

LIVE DEMO

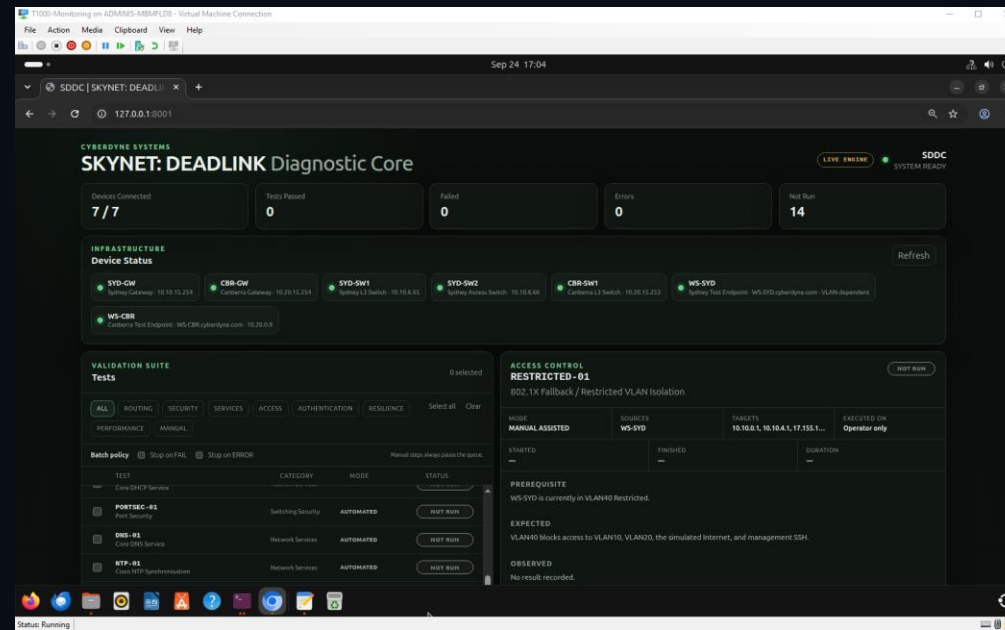




SYSTEM PERFORMANCE

MONITORING, TESTING & BENCHMARKING (REQ. 11)

LIVE DEMO





STANDARD OPERATING PROCEDURES

DOCUMENTATION - TO BE SUBMITTED (REQ. 12)

STRATEGIC REPORT AND ACTION PLAN

Connects high-level organizational goals with measurable tasks by outlining long-term vision, current realities, and step-by-step execution phases

Classified Document Reference: CS-NET-DOC-2029.

PROJECT REPORT

Project goals, current status, budget, and results to keep stakeholders informed. Includes attachments for configurations, IP addressing, topologies and presentation handouts.

Level 5 Clearance required for operational retrieval.

NETWORK DOCUMENTATION

Demonstrates Test results that align with requirements, difficulties associated with the 802.1x implementation and choice of encryption level

Classified Document Reference: CS-NET-DOC-1867.

PERSONAL REFLECTION / JOURNALS

A reflective personal journal of team's experience throughout the project.

Level 9 Clearance required for operational retrieval.

**END OF
PRESENTATION**

